

Ultra-low energy ion irradiation of few-layer MoS₂: defect production and intercalation

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Two-dimensional (2D) materials such as molybdenum disulfide (MoS₂) and other transition-metal dichalcogenides (TMDs) have attracted significant interest due to their unique electronic, optical, and mechanical properties [1]. These materials show great potential for applications in electronics, optoelectronics, catalysis, and energy storage. However, their applications often require wafer-scale manufacturing going beyond current growth-based capabilities. Ion implantation is a powerful approach for top-down modification of materials by introducing controlled defects and dopants [2]. However, the reduced dimensionality of 2D materials poses restrictions for their nanofabrication and requires ion beams to operate at very low energies, from some hundreds of eV to a few tens of eV. The ultra-low energy manufacturing of 2D TMDs is not yet fully understood. In this work, the effects of ultra-low-energy argon and xenon ion implantation on MoS₂ are studied using scanning transmission electron microscopy, Raman spectroscopy, and molecular dynamics simulations. Mechanically exfoliated MoS₂ samples were irradiated with the KIIA ion accelerator at the University of Helsinki at doses of 1×10^{14} to 1×10^{15} ions/cm² with ion energies ranging from 15 to 200 eV. Structural analysis shows pore-type defect formation, while MD simulations reveal that the onset of defect generation in MoS₂ begins at $\sim 35\text{--}40$ eV for Ar, with sulfur sputtering exhibiting a stepwise increase above $\sim 1 \times 10^{14}$ cm⁻², while at lower fluences the increase is more gradual; a similar but less pronounced trend is observed for Mo. These non-linear sputtering effects are more pronounced for Xe and emerge at lower ion energies compared to Ar, intensifying as the ion energy increases. At higher energies, reaching 60 eV for Xe, both S and Mo sputtering rise rapidly around $\sim 5 \times 10^{14}$ cm⁻², indicating accelerated defect accumulation and membrane de-stabilization. This behavior is consistent with previous Ga ion irradiation studies of MoS₂, where low fluences primarily increased pore density while higher fluences ($\sim 10^{14}$ cm⁻²) led to vacancy coalescence and growth of larger nm-scale pores [3]. The simulations further reveal an intermediate energy window where ions can be trapped between MoS₂ layers. However, at high ion

fluence, the trapped ions may escape through nm-sized pores that form when defects coalesce.

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